Rate-Limit Forecast Model

claumon `internal/forecast`

Model version **v1.2** — 2026-05-30

This document specifies the math behind `internal/forecast`. It is normative: implementations must match the formulas here, and deviations are bugs unless this document is updated. The model version above matches `forecast.ModelVersion` in the Go code; retired model versions are preserved under `internal/forecast/archive/vN.M/`, and `CHANGELOG.md` summarises each bump.

1 What we forecast

For each open rate-limit window (session, weekly, per-model weekly), the forecaster reads the time series of utilization snapshots and produces:

1. A point estimate F of the utilization at reset.
2. An 80% **credible interval** $[F^-, F^+]$ for F. (We use the looser "confidence" wording in user-facing strings only; the posterior is Bayesian, see §5.)
3. For each threshold C_{thr} (e.g. 100%), either a median ETA with an 80% CI, or `nil` if the threshold is unlikely to be reached before reset.

This procedure is an *empirical Bayes* forecast: the prior on the rate r and the path-noise variance σ^2 are both estimated from past data (§5, §7) and then plugged into the posterior update, rather than being themselves marginalized under a hyperprior. The conjugate update on r and the Monte Carlo over rate-and-path uncertainty are genuinely Bayesian; the "empirical" qualifier refers to the prior, not the inference step.

A separate forecast is produced per gauge. The only cross-window sharing is through the calibration constants (§7).

2 The model in one paragraph

Inside the current window, utilization $u(t)$ grows at an unknown constant rate r plus Brownian noise (random behavioral pivots). We estimate r from the recent slope of the snapshots (OLS) and from a Gaussian prior fit on past sessions, combined by conjugate update. The forecast at reset is the sum of two random pieces: deterministic-time-times-uncertain-rate, plus Brownian accumulation over the remaining horizon. Both pieces are Gaussian, so the forecast is Gaussian and the CI is a z-quantile. The ETA has no closed form, so we Monte Carlo it.

3 Generative model and assumptions

For $s \in [0, \Delta t_{\text{rem}}]$ with $\Delta t_{\text{rem}} = T_{\text{reset}} - t_{\text{now}}$,

$$u(t_{\text{now}} + s) = u_{\text{now}} + r\,s + W(s), \qquad W(s) \sim \mathcal{N}\big(0,\, \sigma^2_{\text{session}}\, s\big),$$

with r unknown and W a Brownian motion independent of r. Three assumptions:

1. **Linear conditional mean.** r is constant within a window. Violated by abrupt mid-window mode shifts; partially absorbed by the recency window in §5.

2. **Brownian path noise.** Variance grows linearly with time. Approximately correct if behavioral pivots are mean-zero and not too heavy-tailed.

3. **Exchangeable past sessions.** Historical sessions are iid draws from the same noise process. Required for the calibration in §7.

On using Brownian motion for a monotone quantity. Real utilization is monotone non-decreasing in $[0, 1]$: tokens accumulate, the gauge only resets at the window boundary. A Brownian path can drift below u_{now}, above 1, or below 0, none of which can happen physically. We keep BM anyway because it yields a Gaussian forecast and closed-form z-quantile CIs; the principled alternative is a non-decreasing process (Gamma process: $u(t+s) - u_{\text{now}} \sim \text{Gamma}(rs/\beta, \beta)$, or compound Poisson over individual API calls), at the cost of MC for the CI.

The simplification has three concrete consequences:

- **Unphysical lower tail.** $P\big(u(t_{\text{now}} + s) < u_{\text{now}}\big) = \Phi(-r\sqrt{s}/\sigma_{\text{session}})$. With representative numbers ($r = 0.06\,\text{h}^{-1}$, $\sigma^2_{\text{session}} = 2.5 \times 10^{-3}$) this is $\sim 2\%$ at $s = 3\,\text{h}$ but $\sim 20\%$ at $s = 30\,\text{min}$. The display clip at $[u_{\text{now}}, 1]$ (§7) hides this in user-facing CIs, but the moments are biased: the point forecast $F = u_{\text{now}} + \hat{r}_{\text{post}}\,\Delta t_{\text{rem}}$ is *slightly low* relative to the truncated-Gaussian mean, with the bias worst at short horizons where the lower-tail mass is largest.

- **MC ETAs biased late.** First-passage times in §6 are computed on raw BM paths that can dip below threshold and cross later. A monotone process would cross sooner on those paths, so the reported median ETA is conservatively late. Magnitude is bounded by the same lower-tail probability above.

- **Reinterpretation as forecast-error noise.** A defensible reading is that $W(s)$ models the deviation of the realized trajectory from the deterministic prediction $u_{\text{now}} + rs$, not literal token movement. Under that reading $W(s) < 0$ means "the user slowed down relative to recent rate", not "tokens went down". This is consistent with how the calibration in §7 fits $\hat{a}$: against squared forecast errors, not squared step-to-step increments.

The Brownian contribution to σ^2_F is $\Delta t_{\text{rem}}\,\sigma^2_{\text{session}}$ (linear), while the rate-uncertainty contribution from v1.1 is $\Delta t^2_{\text{rem}} \cdot \max(\tau^2_{\text{post}}, \bar{\tau}^2)$ (quadratic), so at long horizons BM is subdominant. If the new `UnderspreadX` diagnostic remains miscalibrated after v1.1 telemetry stabilises, replacing BM with a Gamma or compound-Poisson process is the natural next investigation.

Visualisation note. The forecast trajectory modal in the web UI draws the raw MC paths from §6, which inherit the BM non-monotonicity and visibly dip. The dips disappear once a monotone process replaces BM.

4 Notation

Symbol	Meaning
$u(t)$, u_{now}	utilization at time t and at now, $\in [0, 1]$
t_{now}, T_{reset}, Δt_{rem}	now, reset, remaining horizon (h)
r	true mean growth rate within window (h^{-1})
$\hat{r}_{\text{OLS}}$, SE_{OLS}	OLS estimate and its standard error
μ_0, τ_0^2	Gaussian prior on r from history
$\hat{r}_{\text{post}}$, τ^2_{post}	posterior on r after the OLS update
$\sigma^2_{\text{session}}$	Brownian diffusion coefficient (h^{-1})
F, σ_F^2	point forecast and its variance
$z_{0.9}$	$\Phi^{-1}(0.9) \approx 1.2816$
K, Δs	MC trajectories, step size (defaults: 500, 5 min)

5 Estimating the rate

Combine two sources of information about r.

From the current window. Let $\mathcal{R}$ be the snapshots in $[t_{\text{now}} - \tau_{\text{recent}}, t_{\text{now}}]$ and $n = |\mathcal{R}|$. For $n \geq 3$, fit $u_i = \alpha + r\, t_i + \epsilon_i$ by OLS:

$$\hat{r}_{\text{OLS}} = \frac{S_{tu}}{S_{tt}}, \qquad \text{SE}^2_{\text{OLS}} = \frac{\hat{\sigma}^2_\epsilon}{S_{tt}},$$

with $S_{tt} = \sum(t_i - \bar{t})^2$, $S_{tu} = \sum(t_i - \bar{t})(u_i - \bar{u})$, and $\hat{\sigma}^2_\epsilon = \frac{1}{n-2}\sum(u_i - \hat{\alpha} - \hat{r}_{\text{OLS}}\, t_i)^2$. Default $\tau_{\text{recent}} = 30\,\text{min}$.

Strictly, Brownian residuals are heteroskedastic and serially correlated, so the iid-OLS variance formula above is an approximation. For the short recency window the discrepancy is small and the point estimate is unbiased either way; we accept the approximation and treat SE^2_{OLS} as the likelihood precision in the conjugate update below.

From history. For each past completed session s with final value u_s^* and duration D_s, define $\rho_s = u_s^*/D_s$. The prior mean is the sample mean of ρ_s. For the prior variance, note that under the generative model

$$\rho_s = r_s + \frac{W_s(D_s)}{D_s}, \qquad \text{Var}[\rho_s] = \text{Var}[r_s] + \frac{\sigma^2_{\text{session}}}{D_s},$$

so the raw sample variance of ρ_s overstates $\text{Var}[r_s]$ by the average path-noise contribution. Subtract it off:

$$\mu_0 = \text{mean}_s\, \rho_s, \qquad \tau_0^2 = \max\Big(\text{var}_s\, \rho_s - \sigma^2_{\text{session}} \cdot \text{mean}_s(1/D_s),\ \varepsilon\Big),$$

with $\varepsilon = 10^{-6}$ guarding against negative estimates from small samples. This uses the most recent $\sigma^2_{\text{session}}$ from §7; the two fits depend on each other but only loosely, and the daily refit cycle converges quickly. On the very first fit (before §7 has run), $\sigma^2_{\text{session}} = 0$ and the correction is a no-op.

Fit at startup, refresh daily.

Combine. Normal-normal conjugacy gives the posterior $r \mid \hat{r}_{\text{OLS}} \sim \mathcal{N}(\hat{r}_{\text{post}},\, \tau^2_{\text{post}})$:

$$\frac{1}{\tau^2_{\text{post}}} = \frac{1}{\tau_0^2} + \frac{1}{\text{SE}^2_{\text{OLS}}}, \qquad \hat{r}_{\text{post}} = \tau^2_{\text{post}} \left(\frac{\mu_0}{\tau_0^2} + \frac{\hat{r}_{\text{OLS}}}{\text{SE}^2_{\text{OLS}}} \right).$$

Limits: prior dominates when SE_{OLS} is large (early window); data dominates when SE_{OLS} is small (later in the window). For $n < 3$ the OLS step is undefined; use $\hat{r}_{\text{post}} = \mu_0$, $\tau^2_{\text{post}} = \tau_0^2$.

6 Forecast distribution at reset

The point forecast is

$$F = u_{\text{now}} + \hat{r}_{\text{post}}\, \Delta t_{\text{rem}}.$$

By the law of total variance, conditioning on r:

$$\sigma_F^2 = \underbrace{\Delta t^2_{\text{rem}} \cdot \max(\tau^2_{\text{post}},\, \bar{\tau}^2)}_{\text{rate uncertainty}} + \underbrace{\Delta t_{\text{rem}}\, \sigma^2_{\text{session}}}_{\text{path noise}},$$

where $\bar{\tau}^2$ is the historical-average rate variance recovered in §7 (the quadratic coefficient of the §5 regression). The floor $\max(\tau^2_{\text{post}},\, \bar{\tau}^2)$ is new in v1.1 and corrects a systematic under-spread when assumption A1 (constant within-window rate) is violated: the conjugate τ^2_{post} then accurately estimates uncertainty about the *recent* slope but understates uncertainty about the *end-of-session* slope, because the user pivots. The historical $\bar{\tau}^2$, fit from past end-of-session errors, is the right scale for the latter. We take the max rather than always using $\bar{\tau}^2$ because for short windows where τ^2_{post} is large (few snapshots) the conjugate value should win.

Status as a posterior. The floor is a robustness correction rather than a Bayesian update: $\max(\tau^2_{\text{post}},\, \bar{\tau}^2)$ is not the variance of any coherent posterior over r, so σ_F^2 should be read as v1.0's predictive variance with τ^2_{post} replaced by an empirical lower bound, not as the variance of a posterior predictive. The marginal forecast is still Gaussian (we draw r from a Gaussian, then add Gaussian path noise), with two contributions at different scales: rate uncertainty is quadratic in horizon (dominates at long horizons), path noise is linear (dominates at short).

80% CI. $[F - z_{0.9}\, \sigma_F,\ F + z_{0.9}\, \sigma_F]$, clipped to $[u_{\text{now}}, 1]$ for display: utilization within a reset window is monotone non-decreasing, so the Gaussian left tail dipping below u_{now} is unphysical and the displayed lower bound is floored at u_{now}. The unclipped F is retained for ETA computation.

7 Calibration of $\sigma^2_{\text{session}}$

What we learn. One scalar: how much utilization can drift from the trend per hour of waiting. It is the only quantity history contributes to the forecast *spread* (the prior μ_0, τ_0^2 contributes to the *mean*).

Strategy. Run the forecaster on past sessions, where we already know the answer, and measure how wrong it was as a function of horizon. Read $\sigma^2_{\text{session}}$ off the empirical error structure.

Replay. For each past session s with reset value u_s^* at T_s, sample forecast points $t_f \in [t_s + \tau_{\text{recent}},\, T_s - \Delta_{\text{min}}]$ (default 6 per session, $\Delta_{\text{min}} = 30\,\text{min}$). At each t_f, run §5 on snapshots in $[t_f - \tau_{\text{recent}},\, t_f]$ to obtain $\hat{r}_{\text{post}}(t_f)$, and the projection $\hat{F}(t_f) = u(t_f) + \hat{r}_{\text{post}}(t_f)\,(T_s - t_f)$. Record

$$e_f = u_s^* - \hat{F}(t_f), \qquad \delta_f = T_s - t_f.$$

Two sources of error in each residual. The residual e_f mixes (A) the rate estimate being off at t_f, with the error compounded over δ_f, and (B) Brownian path noise accumulating over δ_f. By the same decomposition as §5,

$$\mathbb{E}[e_f^2 \mid \delta_f] = \underbrace{\delta_f\,\sigma^2_{\text{session}}}_{\text{path noise (B)}} + \underbrace{\delta_f^2\,\bar{\tau}^2}_{\text{rate uncertainty (A)}},$$

with $\bar{\tau}^2$ the average posterior rate variance at the forecast points. The two contributions scale differently in δ_f, which is what lets us separate them. Naive averaging like $\sigma^2_{\text{session}} \approx \text{mean}_f(e_f^2/\delta_f)$ would not separate them and would bias the estimate upward by a $\delta_f\,\bar{\tau}^2$ contamination.

Joint regression (v1.2: weighted). Fit both unknowns by *weighted* least squares of e_f^2 on $[\delta_f,\ \delta_f^2]$ (no intercept), with weights $w_f = 1/\delta_f^2$:

$$(\hat{a}, \hat{b}) = \arg\min_{a,b} \sum_f w_f \big(e_f^2 - a\,\delta_f - b\,\delta_f^2\big)^2, \qquad \sigma^2_{\text{session}} := \max(\hat{a},\ 10^{-6}).$$

The coefficient $\hat{a}$ on the linear term is the per-hour path-noise variance: that is the quantity we want. The weighting corrects heteroskedasticity: e_f^2 is a squared error whose own variance scales as $(\mathbb{E}[e_f^2])^2$ and so grows steeply with δ_f, so the unweighted OLS of v1.0–v1.1 was dominated by the few long-horizon points and inflated $\hat{a}$ — over-predicting short-horizon spread by roughly an order of magnitude, and (through the noise correction in §5) over-subtracting until τ_0^2 floored to ε, which silently pinned the rate prior. Weighting by $1/\delta_f^2$ is a fixed proxy for $1/\text{Var}[e_f^2]$ that puts short- and long-horizon residuals on a comparable scale; the quadratic term $\hat{b}$ stays identified because only the long-horizon points carry information about δ_f^2.

Note that this regression treats $\bar{\tau}^2$ as constant across forecast points, while in reality $\tau^2_{\text{post}}(t_f)$ varies (it depends on how many recent snapshots existed at t_f). If $\tau^2_{\text{post}}(t_f)$ correlates with δ_f — and it usually does, since δ_f is anti-correlated with elapsed time in the session — the missing covariance leaks into $\hat{a}$. A tighter alternative is to subtract the known $\delta_f^2\,\tau^2_{\text{post}}(t_f)$ piece as an offset and regress the residual on δ_f alone; we keep the joint regression for simplicity and absorb the bias into the daily refit.

Role of $\hat{b}$ (v1.1: kept as a floor). Conceptually $\hat{b}$ estimates the same quantity as τ^2_{post} in §5 (the rate variance relevant to predicting $u(T_{\text{reset}})$), just as a historical average instead of a fresh per-forecast value. The original v1.0 design discarded $\hat{b}$ on the grounds that the per-forecast τ^2_{post} uses today's actual snapshots and is therefore strictly more informative.

In practice, on real sessions, τ^2_{post} shrinks to one or two orders of magnitude below $\hat{b}$ within a few snapshots of OLS. Empirically, the $\hat{b}$-sized variance is the one that matches the realized end-of-session errors. The reason is assumption A1: τ^2_{post} measures uncertainty in the *recent* slope, which OLS can pin down precisely; but if the rate shifts mid-window, the slope governing the rest of the session has variance closer to the cross-session $\bar{\tau}^2$ than to today's τ^2_{post}.

v1.1 therefore keeps $\hat{b}$ and stores it as $\bar{\tau}^2$ on the calibration. §6 uses $\max(\tau^2_{\text{post}},\ \bar{\tau}^2)$ in σ_F^2. The MC in §6 likewise samples r_k with standard deviation $\sqrt{\max(\tau^2_{\text{post}},\ \bar{\tau}^2)}$.

The $b\,\delta_f^2$ term has to be present in the regression in any case: without it, the $\delta_f^2\,\bar{\tau}^2$ piece of e_f^2 has nowhere to go but $\hat{a}$, biasing it upward.

Floor and refit. The floor on $\hat{a}$ guards against negative estimates from small samples (OLS does not enforce sign). Refit daily.

8 Threshold ETA

For threshold C_{thr} with $u_{\text{now}} < C_{\text{thr}} \le 1$, the first-passage time is $T^* = \inf\{t > t_{\text{now}} : u(t) \ge C_{\text{thr}}\}$, with $T^* = \infty$ if no crossing in $[t_{\text{now}}, T_{\text{reset}}]$.

Deterministic ETA (point estimate). Setting $W \equiv 0$ and $r = \hat{r}_{\text{post}}$:

$$\tilde{T}^* = t_{\text{now}} + \frac{C_{\text{thr}} - u_{\text{now}}}{\hat{r}_{\text{post}}},$$

defined when $\hat{r}_{\text{post}} > 0$ and $\tilde{T}^* \le T_{\text{reset}}$; else `nil`. Note that $\tilde{T}^*$ is **not** the median of T^* under the full stochastic model. Two effects pull in opposite directions:

- *BM skew (pulls median down).* For BM with fixed drift $\mu > 0$ hitting $a > 0$, $T^* \sim \mathrm{IG}(a/\mu,\, a^2/\sigma^2)$ with mean a/μ and right skew, so $\mathrm{median}(T^* \mid r{=}\mu) < a/\mu$.
- *Rate uncertainty (pulls median up).* Mixing over $r \sim \mathcal{N}(\hat{r}_{\text{post}},\, \tau^2_{\text{post}})$ inflates first-passage times asymmetrically: the map $r \mapsto a/r$ is convex on $r > 0$ and steepens sharply as $r \to 0^+$, so the right tail of T^* stretches far more than the left tail compresses. Any mass at $r \le 0$ contributes additional atoms at $T^* = \infty$. Both effects pull the median of T^* above $\tilde{T}^*$.

Which effect wins depends on the regime, so $\mathrm{median}(T^*)$ can sit on either side of $\tilde{T}^*$. The MC percentiles below are the authoritative summary; $\tilde{T}^*$ is a cheap deterministic anchor, not an estimator of the median.

CI via Monte Carlo. The first-passage time of a Brownian motion with random Gaussian drift has no tractable closed form. Simulate $K = 500$ trajectories:

```
tau_eff^2 = max(tau_post^2, bar_tau^2)     # v1.1 floor; see §4
for k in 1..K:
    r_k    = N(r_hat_post, tau_eff^2)      # may be negative; do NOT clip
    u_k[0] = u_now
    for j in 1..M, dt = step size of jth bucket:
        u_k[j] = u_k[j-1] + r_k * dt + N(0, sigma_session^2 * dt)
    T*_k = linear interpolation of first j with u_k[j] >= C_thr
         = infinity if none
```

Negative r_k draws are kept as-is: those trajectories drift downward and typically yield $T^*_k = \infty$, which is the correct outcome under the model. Clipping to zero would bias crossings upward and contradict §5.

Reported summaries. Let p_∞ be the fraction of trajectories with $T^*_k = \infty$. Define $\tilde{T}^*_{\text{MC}} = \mathrm{median}(T^*_k)$ over the **full** sample (treating ∞ as larger than any finite value):

- If $p_\infty \ge 0.5$: the true median is ∞. Report ETA as `nil`.
- If $p_\infty < 0.5$ but $p_\infty \ge 0.1$: the median is finite but the 90th percentile is ∞. Report median, lower bound = 10th percentile of finite T^*_k, upper bound = `nil` (open-ended).
- If $p_\infty < 0.1$: report median and full 80% CI from the 10th and 90th percentiles of finite T^*_k.

RNG seed derived from $(t_{\text{now}},\, T_{\text{reset}},\, u_{\text{now}},\, \hat{r}_{\text{post}},\, \tau^2_{\text{post}},\, \sigma^2_{\text{session}})$ for test reproducibility.

9 Edge cases

Case	Handling
$n < 3$ snapshots in $\mathcal{R}$	Use prior alone: $\hat{r}_{\text{post}} = \mu_0$, $\tau^2_{\text{post}} = \tau^2_0$
Prior empty ($\lvert\mathcal{S}\rvert < 2$)	Suppress forecast; UI shows "collecting data"
Reset detected in $\mathcal{R}$ ($u_{i+1} < u_i$)	Drop pre-reset snapshots; refit
$C_{\text{thr}} \le u_{\text{now}}$	Threshold already crossed; return $T^* = t_{\text{now}}$
$\hat{r}_{\text{post}} \le 0$	Deterministic ETA undefined; rely on MC summary
$F \notin [0, 1]$	Clip for display; retain unclipped for downstream

10 Worked example

5-hour session window: $t_{\text{start}} = 11{:}00$, $T_{\text{reset}} = 16{:}00$, $t_{\text{now}} = 13{:}00$, $u_{\text{now}} = 0.30$, so $\Delta t_{\text{rem}} = 3\,\text{h}$.

Last four snapshots ($\tau_{\text{recent}} = 30\,\text{min}$):

i	t_i (h from 12:30)	u_i
1	0.000	0.270
2	0.167	0.280
3	0.333	0.295
4	0.500	0.300

OLS: $\hat{r}_{\text{OLS}} \approx 0.0630\,\text{h}^{-1}$, $\text{SE}^2_{\text{OLS}} \approx 6.30 \times 10^{-5}$.

Prior (suppose, from history, after the §5 noise correction): $\mu_0 = 0.080\,\text{h}^{-1}$, $\tau^2_0 = 3.6 \times 10^{-3}$.

Posterior: $\tau^2_{\text{post}} \approx 6.19 \times 10^{-5}$, $\hat{r}_{\text{post}} \approx 0.0633\,\text{h}^{-1}$ (data dominates).

Calibration: $\sigma^2_{\text{session}} = 2.5 \times 10^{-3}\,\text{h}^{-1}$, $\bar{\tau}^2 = 3.6 \times 10^{-3}\,\text{h}^{-2}$ (approximately matches τ^2_0 in a stable regime, as expected).

Forecast:

$$F = 0.30 + 0.0633 \cdot 3 \approx 0.490,$$

$$\max(\tau^2_{\text{post}}, \bar{\tau}^2) = \max(6.19 \times 10^{-5}, 3.6 \times 10^{-3}) = 3.6 \times 10^{-3} \quad \text{(floor active)},$$

$$\sigma^2_F \approx 9 \cdot 3.6 \times 10^{-3} + 3 \cdot 2.5 \times 10^{-3} \approx 3.24 \times 10^{-2} + 7.5 \times 10^{-3} \approx 4.0 \times 10^{-2},$$

$$\sigma_F \approx 0.200.$$

80% CI: $[0.490 - 1.282 \cdot 0.200,\ 0.490 + 1.282 \cdot 0.200] \approx [0.234,\ 0.746]$. Display: "Projected 49% by reset (80% CI: 23%–75%)".

Rate uncertainty dominates path noise by $4.3\times$. The recent slope is locally well-estimated, but historically the rate has varied a lot across sessions, and the v1.1 floor honors that.

ETA to 100%: gap is 0.70, deterministic crossing at $13{:}00 + 0.70/0.0633 \approx 24{:}04$, far beyond reset. Returned as `nil`. The MC confirms by producing infinite crossings for almost all trajectories.

11 Limitations and upgrade paths

- **Heavy-tailed pivots.** Real per-hour increments are heavier-tailed than Gaussian. The 80% CI undercovers in the tails. Cheap fix when calibration data is plentiful: replace $z_{0.9}$ with an empirical quantile of standardized residuals.
- **Constant-rate assumption.** A mode shift within the window (deep work → meetings) breaks (A1). The recency window τ_{recent} absorbs slow drift but not abrupt shifts. The principled upgrade is a Kalman filter where r is a slowly-varying latent process.

- **Global $\sigma^2_{\textbf{session}}$.** Pivot variance may depend on day-of-week or project type. Stratified calibration is the upgrade, but requires more sessions per stratum.